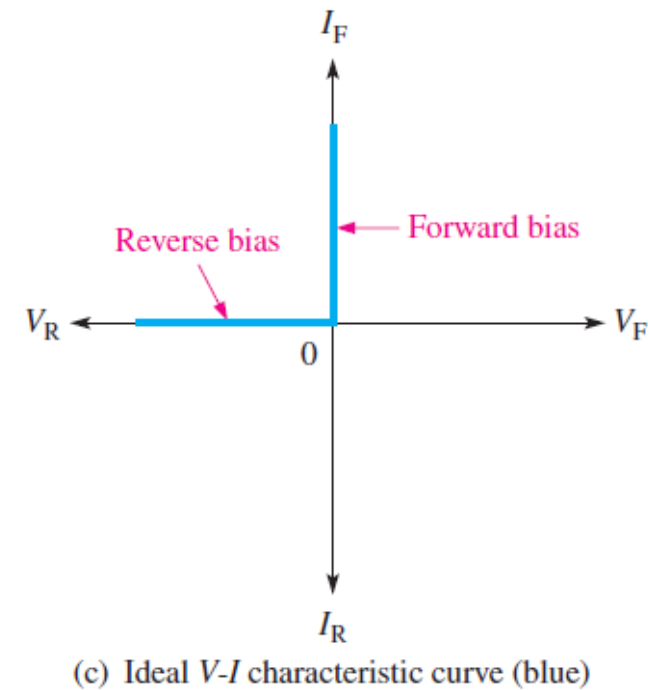
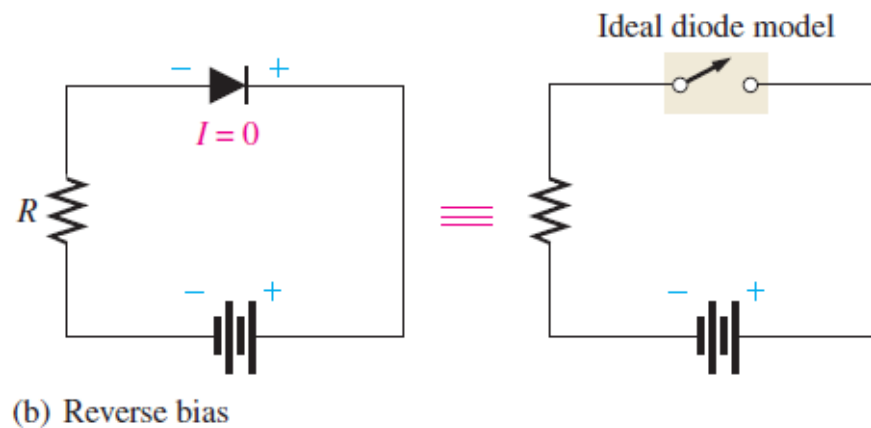
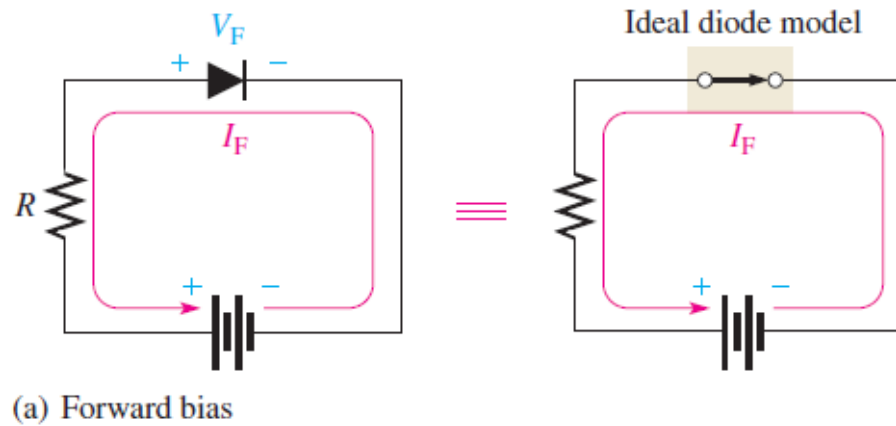


Diode Models

- Ideal Diode Model

- It is the least accurate approximate model
- The diode can be replaced by a simple switch
- When diode is forward bias, diode acts like a closed switch
- When diode is reverse biased, diode acts like an open switch
- The barrier potential, dynamic resistance of diode and reverse current are neglected
- Only used for troubleshooting purpose, whether diode is working or not

Diode Models (contd.)



Diode Models (contd.)

- Since, barrier potential and dynamic resistance is neglected, the voltage across diode in forward bias is zero and current can be calculated as

$$I_F = \frac{V_{Bias}}{R_{Limit}}$$

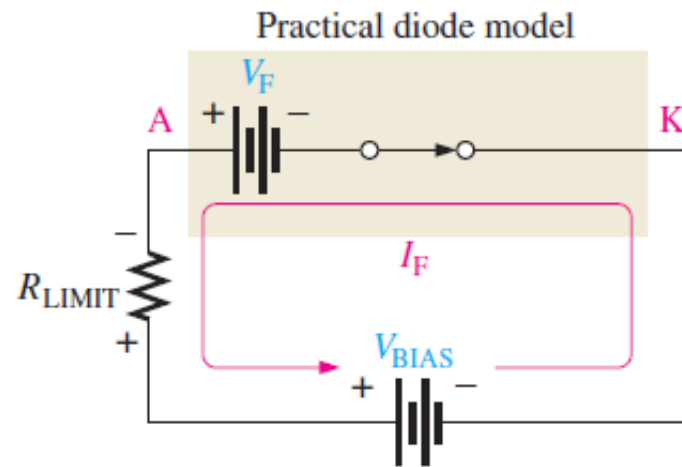
- Since, reverse current is neglected, means reverse current is zero and reverse voltage is equal to the bias voltage

$$I_R = 0 \quad \text{and} \quad V_R = V_{Bias}$$

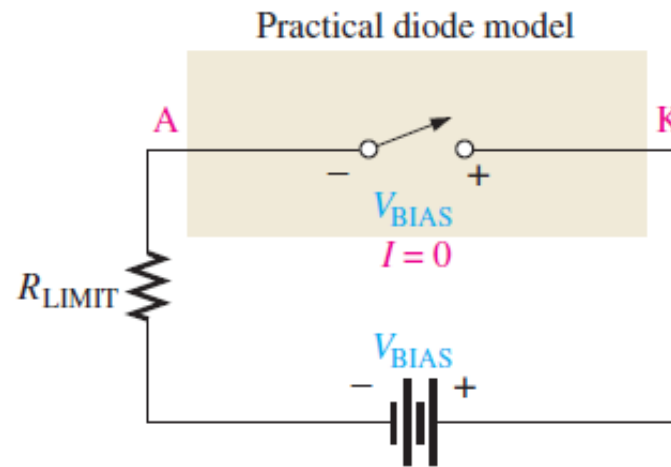
Diode Models (contd.)

- **Practical Diode Model:**
 - In this approximation, the barrier potential is considered i.e. 0.7V for Si
 - In forward bias, a voltage source is considered with a closed switch
 - The +ive side of the source is at anode
 - Bias voltage should be greater than that voltage source in order to conduct a diode
 - In reverse bias, voltage source will not effect the circuit as diode acts as an open switch

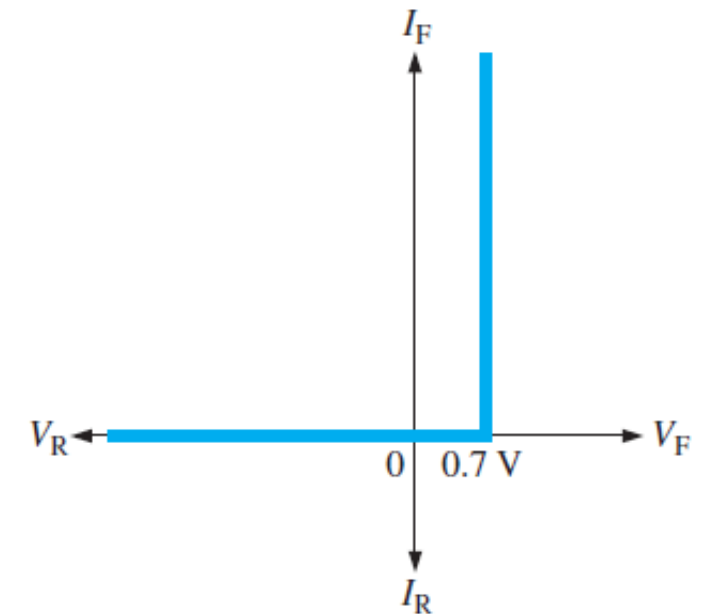
Diode Models (contd.)



(a) Forward bias



(b) Reverse bias



(c) Characteristic curve (silicon)

Diode Models (contd.)

- As diode has a voltage drop of 0.7V so

$$V_F = 0.7V$$

- The current through the diode can be calculated by KCL, hence

$$I_F = \frac{V_{Bias} - V_F}{R_{Limit}}$$

- In reverse bias, reverse current is zero and reverse voltage is equal to the bias voltage

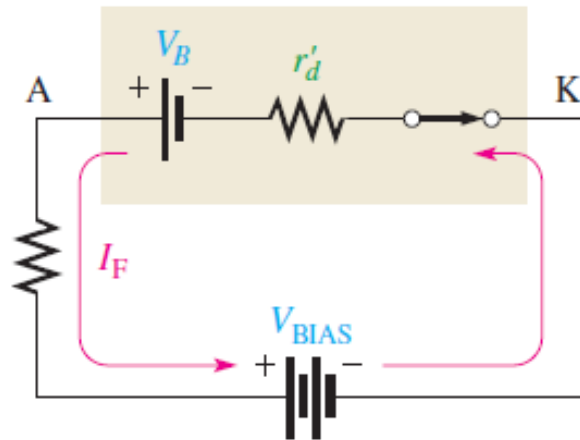
$$I_R = 0 \quad \text{and} \quad V_R = V_{Bias}$$

- This approximation is useful when dealing with the low voltage calculations and designing basic diode circuits

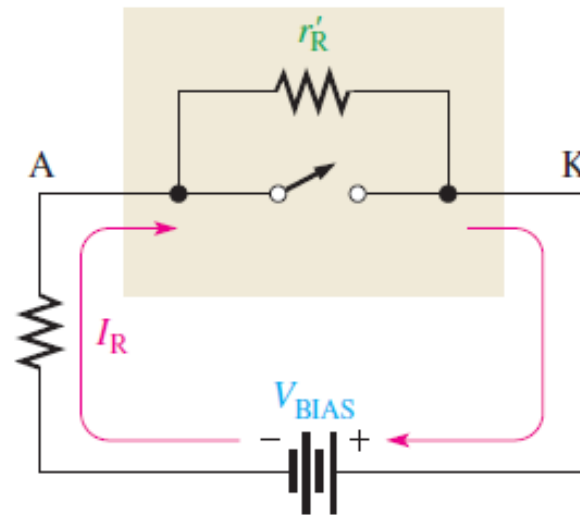
Diode Models (contd.)

- **Complete Diode Model:**
 - It is the most accurate diode approximation
 - It includes the barrier potential, a small forward internal dynamic resistance and a large internal reverse resistance
 - Reverse resistance is taken because it provides a path for reverse current which is included in the approximation

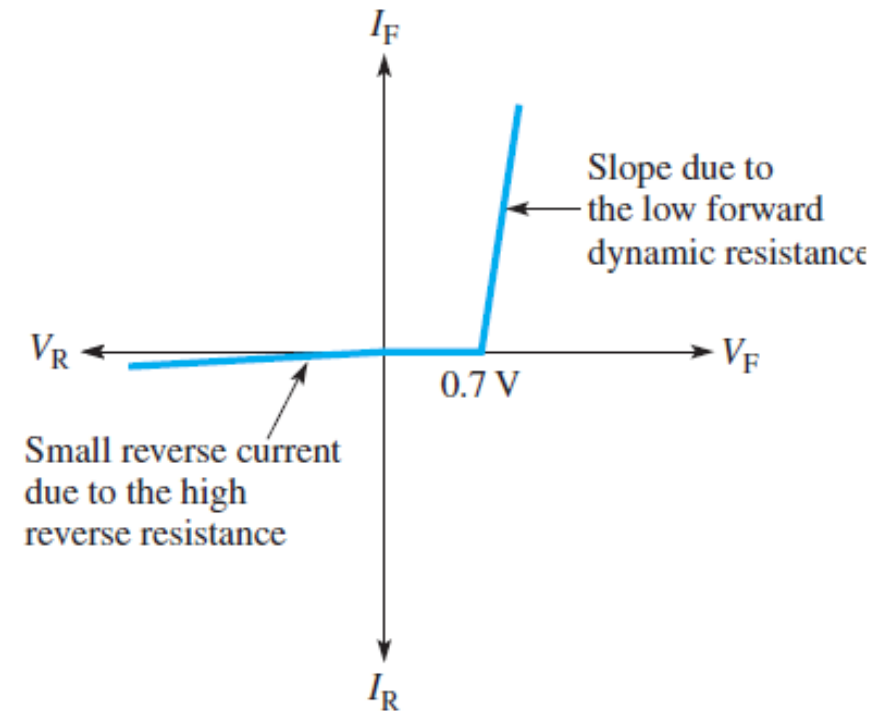
Diode Models (contd.)



(a) Forward bias



(b) Reverse bias



(c) V - I characteristic curve

Diode Models (contd.)

- The values for the forward voltage and current can be calculated as:

$$V_F = 0.7V + I_F r_d$$

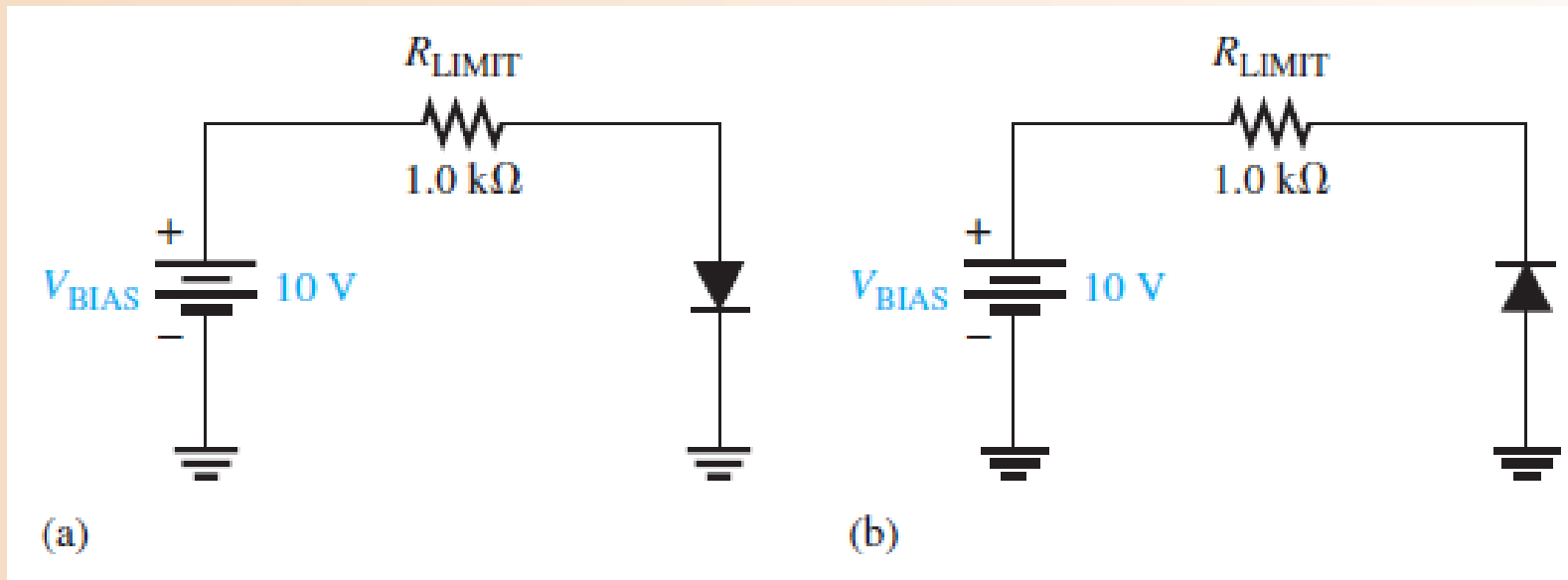
and

$$I_F = \frac{V_{Bias} - 0.7V}{R_{Limit} + r_d}$$

Diode Models (contd.)

Determine the forward voltage and forward current for the diode in Figure for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume $r'_d = 10\ \Omega$ at the determined value of forward current.

Determine the reverse voltage and reverse current for the diode in Figure for each of the diode models. Also find the voltage across the limiting resistor in each case. Assume $I_R = 1\ \mu\text{A}$.



Diode Models (contd.)

Solution (a) Ideal model:

$$V_F = 0 \text{ V}$$

$$I_F = \frac{V_{\text{BIAS}}}{R_{\text{LIMIT}}} = \frac{10 \text{ V}}{1.0 \text{ k}\Omega} = 10 \text{ mA}$$

$$V_{R_{\text{LIMIT}}} = I_F R_{\text{LIMIT}} = (10 \text{ mA})(1.0 \text{ k}\Omega) = 10 \text{ V}$$

Practical model:

$$V_F = 0.7 \text{ V}$$

$$I_F = \frac{V_{\text{BIAS}} - V_F}{R_{\text{LIMIT}}} = \frac{10 \text{ V} - 0.7 \text{ V}}{1.0 \text{ k}\Omega} = \frac{9.3 \text{ V}}{1.0 \text{ k}\Omega} = 9.3 \text{ mA}$$

$$V_{R_{\text{LIMIT}}} = I_F R_{\text{LIMIT}} = (9.3 \text{ mA})(1.0 \text{ k}\Omega) = 9.3 \text{ V}$$

Complete model:

$$I_F = \frac{V_{\text{BIAS}} - 0.7 \text{ V}}{R_{\text{LIMIT}} + r'_d} = \frac{10 \text{ V} - 0.7 \text{ V}}{1.0 \text{ k}\Omega + 10 \Omega} = \frac{9.3 \text{ V}}{1010 \Omega} = 9.21 \text{ mA}$$

$$V_F = 0.7 \text{ V} + I_F r'_d = 0.7 \text{ V} + (9.21 \text{ mA})(10 \Omega) = 792 \text{ mV}$$

$$V_{R_{\text{LIMIT}}} = I_F R_{\text{LIMIT}} = (9.21 \text{ mA})(1.0 \text{ k}\Omega) = 9.21 \text{ V}$$

Diode Models (contd.)

(b) Ideal model:

$$I_R = 0 \text{ A}$$

$$V_R = V_{\text{BIAS}} = 10 \text{ V}$$

$$V_{R_{\text{LIMIT}}} = 0 \text{ V}$$

Practical model:

$$I_R = 0 \text{ A}$$

$$V_R = V_{\text{BIAS}} = 10 \text{ V}$$

$$V_{R_{\text{LIMIT}}} = 0 \text{ V}$$

Complete model:

$$I_R = 1 \mu\text{A}$$

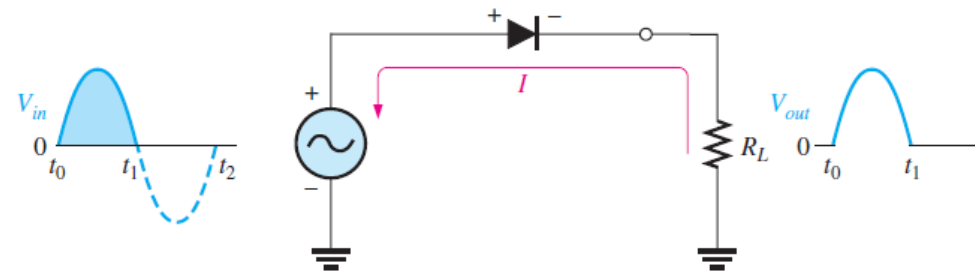
$$V_{R_{\text{LIMIT}}} = I_R R_{\text{LIMIT}} = (1 \mu\text{A})(1.0 \text{ k}\Omega) = 1 \text{ mV}$$

$$V_R = V_{\text{BIAS}} - V_{R_{\text{LIMIT}}} = 10 \text{ V} - 1 \text{ mV} = 9.999 \text{ V}$$

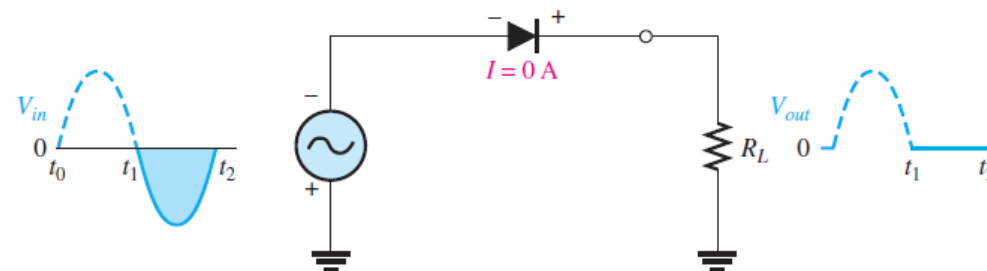
Half Wave Rectifier

- Diodes are mainly used in the power supply circuits
- Power supply converts the standard 230V AC to some DC voltage level
- Main part of a dc supply is the rectifier
- There are two types of rectifiers
 - Half wave rectifier
 - Full wave rectifier

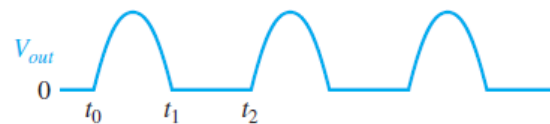
Half Wave Rectifier (contd.)



(a) During the positive alternation of the 60 Hz input voltage, the output voltage looks like the positive half of the input voltage. The current path is through ground back to the source.



(b) During the negative alternation of the input voltage, the current is 0, so the output voltage is also 0.



(c) 60 Hz half-wave output voltage for three input cycles

Half Wave Rectifier (contd.)

- During the positive cycle diode conducts and current flow through the resistor
- For a negative cycle, diode goes into reverse biasing and do not operate, no current flows through resistor
- Net result is that only positive cycle of AC source appears across load resistor
- No polarity change at the output so a pulsating dc voltage appears across diode

Half Wave Rectifier (contd.)

- Average value of half wave output voltage can be calculated as:

$$V_{Avg} = \frac{V_P}{\pi}$$

- Equation shows that V_{avg} is approx. 31.8% of V_p (Ideal Diode Case)

- When a practical model is used peak output voltage can be calculated as:

$$V_{P(out)} = V_{P(in)} - 0.7V$$